

MATH 302: Homework 1

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Question 1

Question. Show that every normed vector space $(V, \|\cdot\|)$ is automatically a metric space (V, d) with the metric d given by $d(x, y) = \|x - y\|$.

Definition (1.1 Metric Space Properties) . A set is a metric space if it is equipped with a function $d : V \times V \rightarrow \mathbb{R}$ such that for all $x, y, z \in V$:

Property	Description
I. Identity of Indiscernibles	$d(x, y) = 0 \Leftrightarrow x = y$
II. Symmetry	$d(x, y) = d(y, x)$
III. Triangle Inequality	$d(x, y) \leq d(x, z) + d(z, y)$
IV. Non-Negativity	$d(x, y) \geq 0$

Let there be a normed vector space $(V, \|\cdot\|)$. Define $d(x, y) = \|x - y\|$. We will show that d satisfies the four properties of a metric space.

Identity of Indiscernibles

Observe that the “Identity of Indiscernibles” property states that $d(x, y) = 0$ if and only if $x = y$. In other words given $x, y \in V$ and the distance between them is zero, then they must be the same point.

By the definition of d , we have:

$$d(x, y) = \|x - y\|$$

By the definition of a norm, $\|v\| = 0$ if and only if $v = 0$. Applying this to $v = x - y$:

$$d(x, y) = 0 \Leftrightarrow \|x - y\| = 0 \Leftrightarrow x - y = 0 \Leftrightarrow x = y,$$

where the final step follows by adding y to both sides. Thus, the “Identity of Indiscernibles” property holds. ■

Symmetry

Observe the “Symmetry” property holds if the distance from x to y is the same as the distance from y to x . In other words, given $x, y \in V$, we have that $d(x, y) = d(y, x)$.

We can test this assertion by substituting the definition of d into the equation:

$$d(x, y) = \|x - y\| = \|-(y - x)\| = \|y - x\| = d(y, x)$$

Thus, the “Symmetry” property holds. ■

Triangle Inequality

Observe the “Triangle Inequality” property states the distance between two points x and y is less than or equal to the sum of the distances from x to a third point z and from z to y . In other words, given $x, y, z \in V$, we have that $d(x, y) \leq d(x, z) + d(z, y)$.

We can test this assertion by substituting the definition of d into the equation:

$$\begin{aligned} d(x, z) &= \|x - z\| \wedge d(z, y) = \|z - y\| \wedge d(x, y) = \|x - y\| \\ d(x, z) + d(z, y) &= \|x - z\| + \|z - y\| \end{aligned}$$

Observe the following:

$$\|x - y\| = \|(x - z) + (z - y)\| \leq \|x - z\| + \|z - y\|$$

Thus, the “Triangle Inequality” property holds. ■

Non-Negativity

Observe the “Non-Negativity” property states that the distance between two points x and y is always greater than or equal to zero. In other words, there can never be a negative distance between two points.

As the distance function d is defined as the norm of the difference between two points, we have that $d(x, y) = \|x - y\|$. By the definition of a norm, we know that $\|v\| \geq 0$ for all $v \in V$. Therefore, $d(x, y) \geq 0$ for all $x, y \in V$. ■

Conclusion

As we have shown that d satisfies all four properties of a metric space, we conclude that every normed vector space $(V, \|\cdot\|)$ is automatically a metric space (V, d) with the metric d given by $d(x, y) = \|x - y\|$.

Question 2

Question. On $\mathbb{R}^n$ define, for any $x = (x_1, \dots, x_n)$,

$$\|x\|_1 = \sum_{j=1}^n |x_j| \quad \text{and} \quad \|x\|_\infty = \max\{|x_1|, \dots, |x_n|\}.$$

Show that both $\|x\|_1$ and $\|x\|_\infty$ are norms on $\mathbb{R}^n$.

Definition (Definition of a Norm) . An element is a norm if the following is met:

Property	Description
I. Positive Definite	$\ x\ = 0 \iff x = 0$
II. Absolute Homogeneity	$\ ax\ = \ a\ \ x\ $
III. Triangle Inequality	$\ x + y\ \leq \ x\ + \ y\ $

WWTS $\|x\|_1$ and $\|x\|_\infty$ maintain these properties on the metric space $\mathbb{R}^n$

Positive Definite

Let $p := (\|x\| = 0)$ and $q := (x = 0)$

Forward Proof

WWTS $p \implies q \equiv \sim p \vee q \equiv (\|x\| \neq 0) \implies (x = 0)$

Case $\|x\|_1$

Consider $\|x\|_1 \neq 0$. Thus it follows by substitution:

$$\|x\|_1 = \sum_{j=1}^n |x_j| = 0$$

As $\forall x, x \in \mathbb{R}$ it follows then $\forall |x|, 0 \leq |x|$ by the properties of absolute values.

WLOG Let $x_a, x_b \in \{x_1, \dots, x_n\}$. Observe the following:

$$\begin{array}{r} 0 < |x_a| \\ + 0 < |x_b| \\ \hline 0 < |x_a| + |x_b| \end{array}$$

Case $\|x\|_\infty$

Consider $\|x\|_\infty \neq 0$. Thus it follows by substitution:

$$\|x\|_\infty = \max\{|x_1|, \dots, |x_n|\} = 0$$

$$\|x\|_\infty \equiv |x_a| : \forall x \in \{|x_1|, \dots, |x_n|\}, x \leq |x_a|$$

By trichotomy $\forall x \in \{x_1, \dots, x_n\}$ we can say x must be either positive, negative, or exactly zero.
Given $|x_a| = 0$

$$|x_a| = 0 \implies x_a = 0$$

Back Proof

WWTS $q \implies p \equiv \sim q \vee p \equiv (x \neq 0) \implies (\|x\| = 0)$